MS1Vtx_maxclusE_raw

MC vs data · overview · rank 29 / 98

rank 29 / 98 by endcap D_equal

used by: NN1 barrel, NN1 endcap

rank-identical columns merged into this one:
    MS1Vtx_barrel_maxclusE
    MS1Vtx_endcap_maxclusE

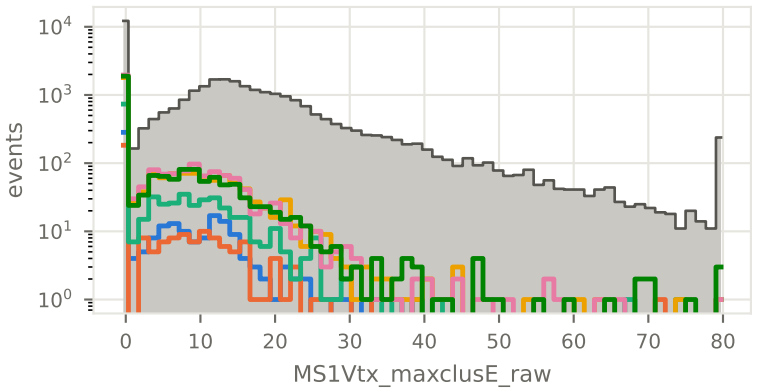
region	D_equal	noise	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg	rank
endcap	0.374	0.022	0.378	0.386	0.351	0.386	0.375	0.376	0.386	34,970	29/98
barrel	0.312	0.051	0.314	0.332	0.275	0.331	0.328	0.319	0.295	1,089	31/98
all	0.411	0.018	0.414	0.419	0.394	0.419	0.414	0.417	0.418	36,059	20/98

MS1Vtx_maxclusE_raw

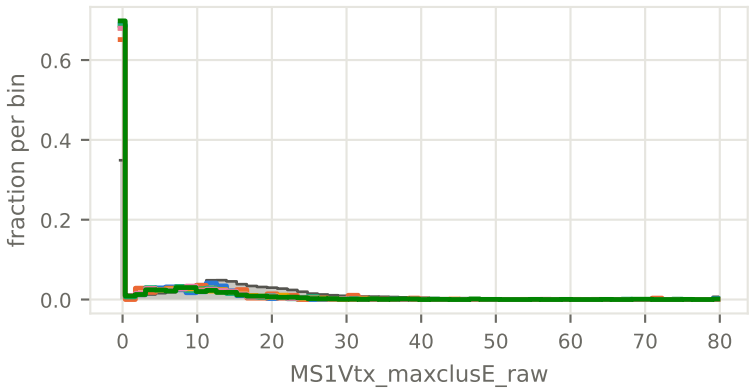
region: ENDCAP · rank 29 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.374 (noise 0.022) D_pooled 0.378 min 0.351 (mH125_mS55) max 0.386 (mH200_mS50) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

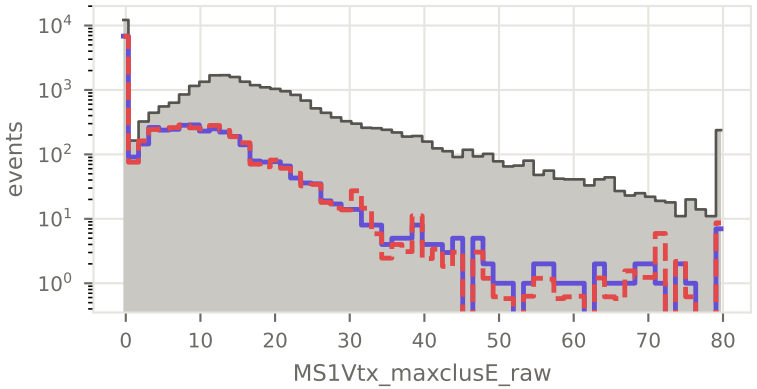


1b · each mass point vs background · normalized

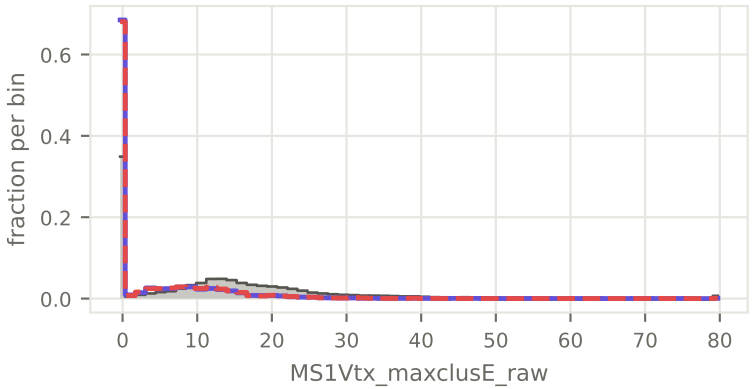


1 · each mass point
background n 34,970
mH125_mS15 D 0.386 n 411
mH125_mS55 D 0.351 n 281
mH200_mS50 D 0.386 n 1,071
mH400_mS100 D 0.375 n 2,648
mH600_mS150 D 0.376 n 2,882
mH1000_mS275 D 0.386 n 2,703

2a · combined signal vs background · raw

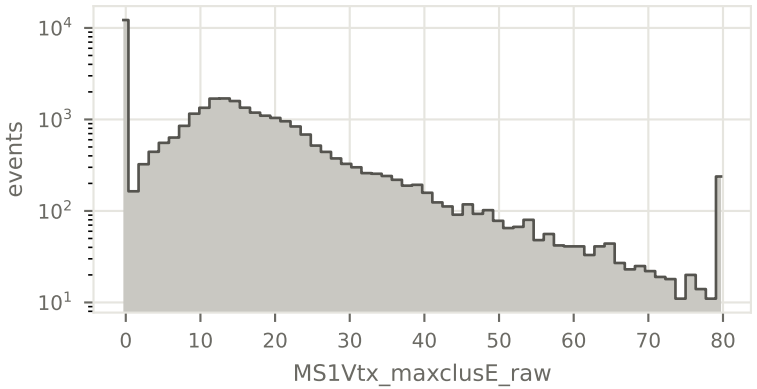


2b · combined signal vs background · normalized

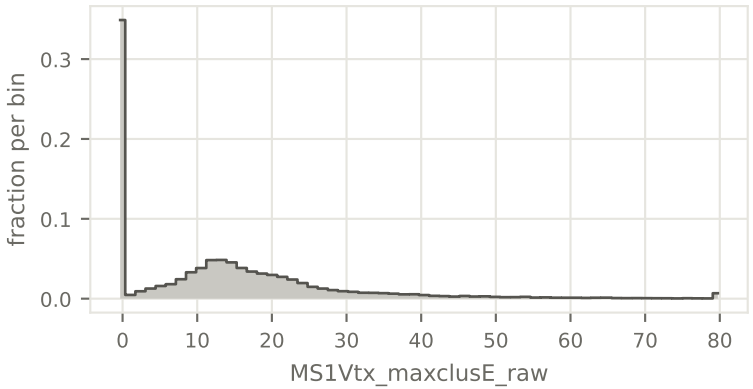


2 · combined signal
background n 34,970
combined, pooled D 0.378 n 9,996
equal weight per mass D 0.374

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

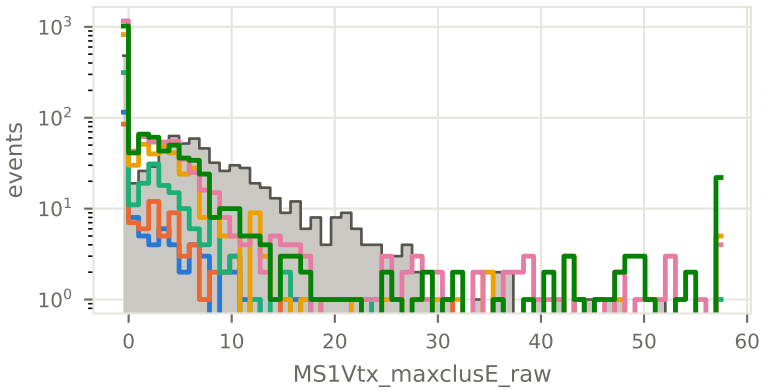
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

MS1Vtx_maxclusE_raw

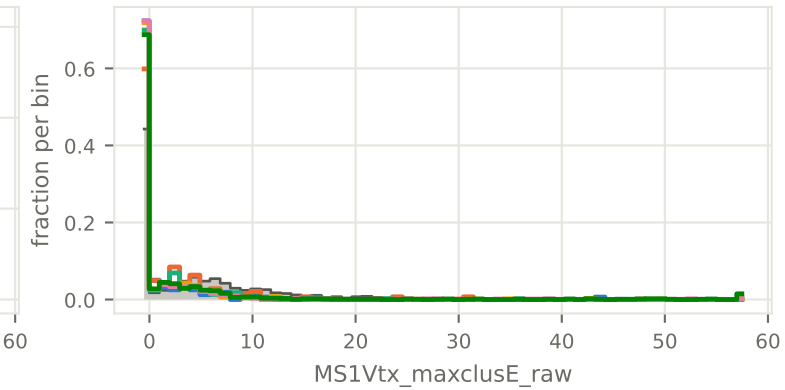
region: BARREL · rank 29 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.312 (noise 0.051) D_pooled 0.314 min 0.275 (mH125_mS55) max 0.332 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

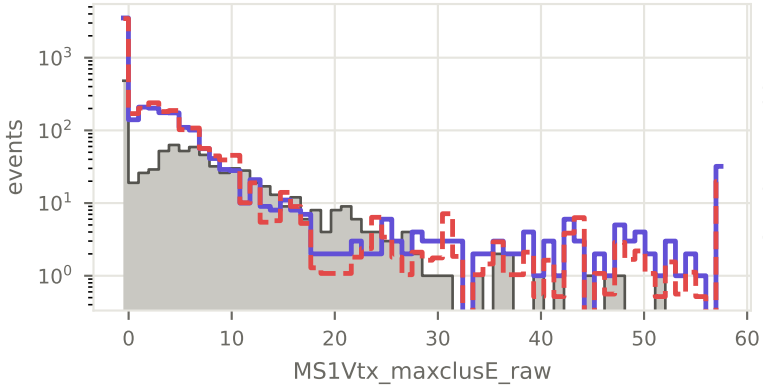


1b · each mass point vs background · normalized

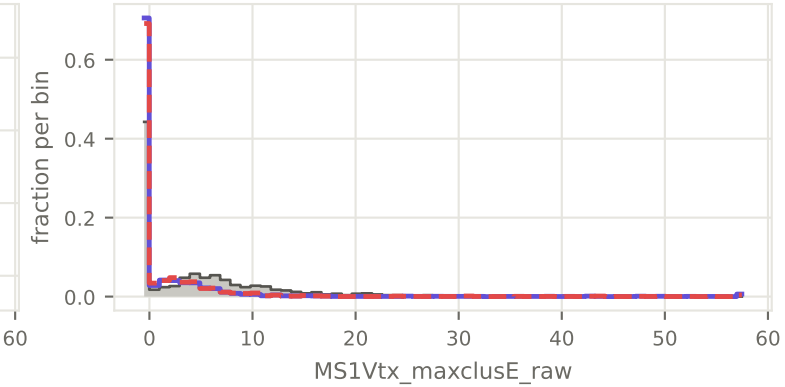


1 · each mass point
background n 1,089
mH125_mS15 D 0.332 n 159
mH125_mS55 D 0.275 n 142
mH200_mS50 D 0.331 n 449
mH400_mS100 D 0.328 n 1,144
mH600_mS150 D 0.319 n 1,608
mH1000_mS275 D 0.295 n 1,485

2a · combined signal vs background · raw

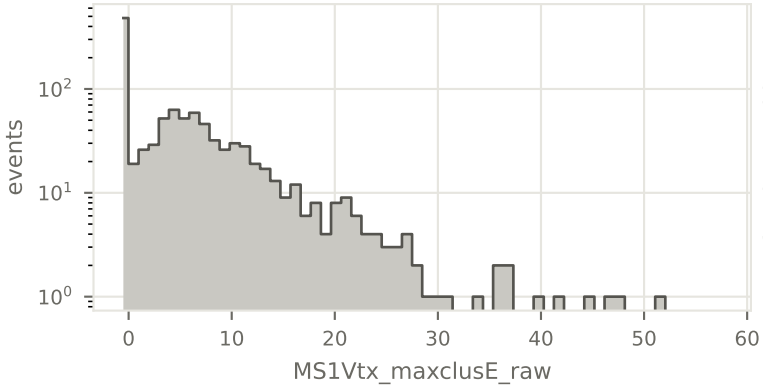


2b · combined signal vs background · normalized

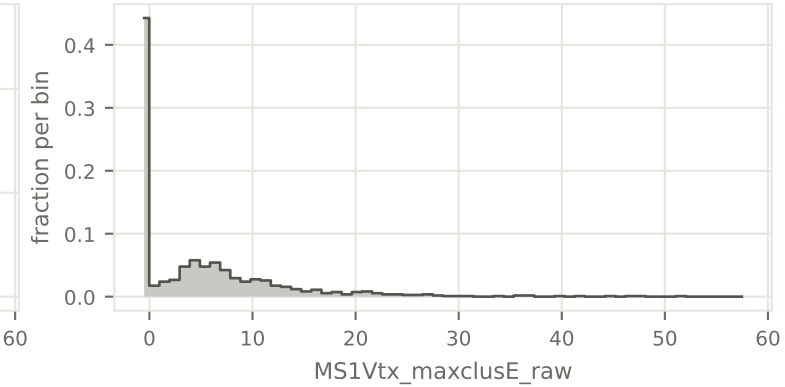


2 · combined signal
background n 1,089
combined, pooled D 0.314 n 4,987
equal weight per mass D 0.312

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

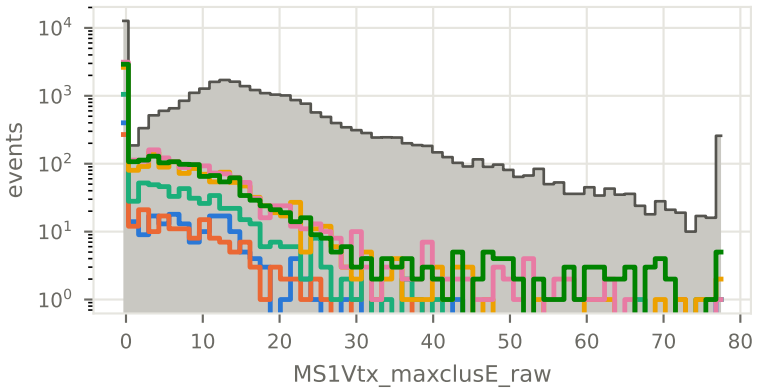
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

MS1Vtx_maxclusE_raw

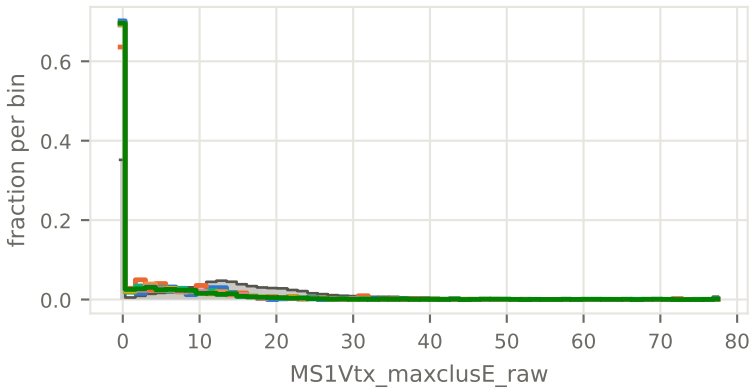
region: ALL · rank 29 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.411 (noise 0.018) D_pooled 0.414 min 0.394 (mH125_mS55) max 0.419 (mH125_mS15) bkg n = 36,059 sig n = 14,983

1a · each mass point vs background · raw

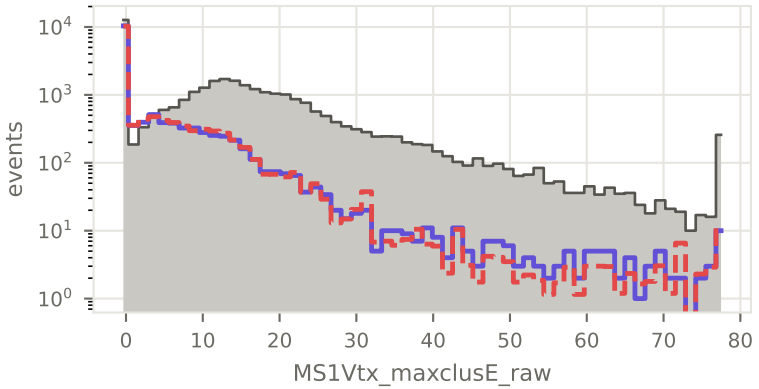


1b · each mass point vs background · normalized

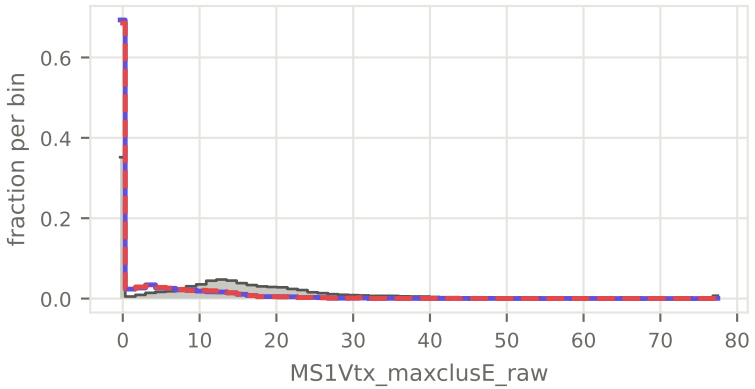


1 · each mass point
background n 36,059
mH125_mS15 D 0.419 n 570
mH125_mS55 D 0.394 n 423
mH200_mS50 D 0.419 n 1,520
mH400_mS100 D 0.414 n 3,792
mH600_mS150 D 0.417 n 4,490
mH1000_mS275 D 0.418 n 4,188

2a · combined signal vs background · raw

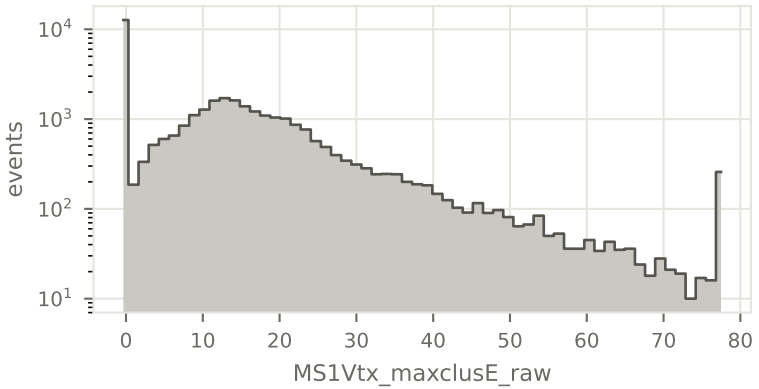


2b · combined signal vs background · normalized

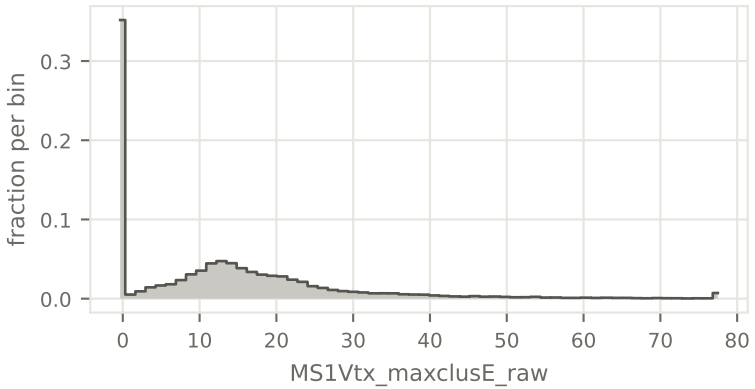


2 · combined signal
background n 36,059
combined, pooled D 0.414 n 14,983
equal weight per mass D 0.411

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 36,059

a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed